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Introduction to C++

Problems to do in Lab. Do not report this.

Problem 1: Sum of Two Numbers

Write a program that takes two integers as input from the user, calculates their sum, and prints the result.

Problem 2: Area of a Rectangle

Write a program that takes the length and width of a rectangle from the user, calculates the area, and prints the result.

Problem 3: Sales Prediction

The East Coast sales division of a company generates 58 percent of total sales. Based on that percentage, write a program that will predict how much the East Coast division will generate if the company has $8.6 million in sales this year.

Problem 4: Restaurant Bill

Write a program that computes the tax and tip on a restaurant bill for a patron with a $88.67 meal charge. The tax should be 6.75 percent of the meal cost. The tip should be 20 percent of the total after adding the tax. Display the meal cost, tax amount, tip amount, and total bill on the screen.

To be reported on canvas. Create a PDF. Include screenshots of code and execution. Include copy-pasteable text of code. Be careful with variable names and indentation. You must use the templates.

Problem 5: Annual Pay

Suppose an employee gets paid every two weeks and earns $2,100 each pay period.

In a year the employee gets paid 26 times. Write a program that defines the following variables:

payAmount. This variable will hold the amount of pay the employee earns each pay period. Initialize the variable with 2100.0.

payPeriods. This variable will hold the number of pay periods in a year. Initialize the variable with 26.

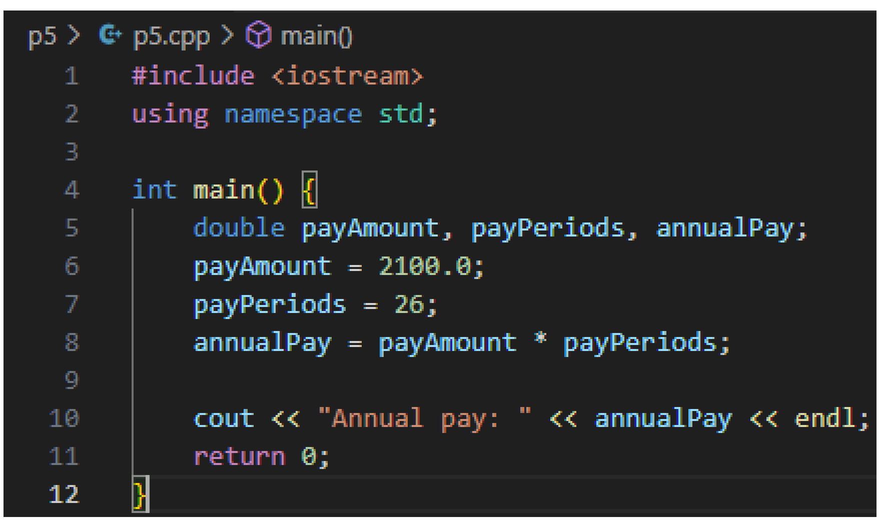
annualPay. This variable will hold the employee's total annual pay, which will be calculated.

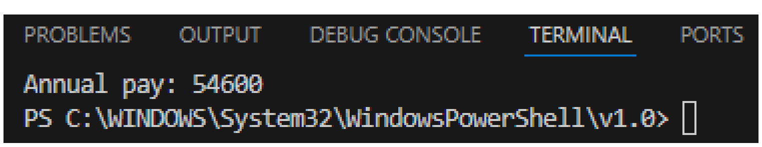
The program should calculate the employee's total annual pay by multiplying the employee's pay amount by the number of pay periods in a year and store the result in the annualPay variable. Display the total annual pay on the screen.

YOU MUST USE NEXT TEMPLATE:

```
//DO NOT MODIFY THIS SECTION
#include <iostream>
using namespace std;
int main()
{
    double payAmount, payPeriods, annualPay;
    payAmount = 2100.0;
    payPeriods = 26;
    annualPay = payAmount * payPeriods;

    cout << "Annual pay: " << annualPay << endl;
    return 0;
}
```





Problem 6: Purchase in a Store

A customer in a store is purchasing four items. The prices of the four items are

Price of item 1 = $15.95

Price of item 2 = $27.95

Price of item 3 = $7.95

Price of item 4 = $12.95

Write a program that holds the prices of the four items in four variables. Display each item's price, the subtotal of the sale, the amount of sales tax, and the total. Assume the sales tax is 7%.

USE THIS TEMPLATE:

```
//DO NOT MODIFY THIS SECTION
#include <iostream>
using namespace std;
int main()
{
    double item1, item2, item3, item4;
    item1 = 15.95;
    item2 = 27.95;
    item3 = 7.95;
    item4 = 12.95;
    const double tax = 0.07;
    double subtotal = 0, total = 0;

    // Add item prices to subtotal
    subtotal+=item1;
    subtotal+=item2;
    subtotal+=item3;
    subtotal+=item4;

    // Display subtotal and tax rate
    cout << "Subtotal:\t" << subtotal << endl;
    cout << "Tax:\t\t" << (tax * 100) << " %" << endl;

    // Add tax to all items
    total += (item1 += (item1 * tax)); // Approx. $17.07
    total += (item2 += (item2 * tax)); // Approx. $29.91
    total += (item3 += (item3 * tax)); // Approx. $8.51
    total += (item4 += (item4 * tax)); // Approx. $13.86
```

```
    // Display total
    cout << "Total:\t\t" << total << endl;
    return 0;
}
```

p6 > p6.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main() {
    // Define variables
    double item1, item2, item3, item4;
    item1 = 15.95;
    item2 = 27.95;
    item3 = 7.95;
    item4 = 12.95;
    const double tax = 0.07;
    double subtotal = 0, total = 0;

    // Add item prices to subtotal
    subtotal+=item1;
    subtotal+=item2;
    subtotal+=item3;
    subtotal+=item4;

    // Display subtotal and tax rate
    cout << "Subtotal:\t" << subtotal << endl;
    cout << "Tax:\t\t" << (tax * 100) << " %" << endl;

    // Add tax to all items
    total += (item1 += (item1 * tax)); // Approx. $17.07
    total += (item2 += (item2 * tax)); // Approx. $29.91
    total += (item3 += (item3 * tax)); // Approx. $8.51
    total += (item4 += (item4 * tax)); // Approx. $13.86

    // Display total
    cout << "Total:\t\t" << total << endl;
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Subtotal:       64.8
Tax:            7 %
Total:          69.336
PS C:\WINDOWS\System32\WindowsPowerShell\v1.0>
```